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40 marks → 40% Mid-sem
60% End-Sem

HS 125

Introductory Macroeconomics

Dornbusch - Fischer & Startz - Macroeconomics)
Mankiw (Macroeconomics by Gregory Mankiw)

Mid Sem
1. National Income Accounting
2. Income Spending Model
3. Money, Interest Rate & Income (IS-LM Model)
4. Monetary Policy, Fiscal Policy
5. International linkages, open economy IS-LM
6. Aggregate supply, Wages, Prices & Unemployment
   - Rational Expectations Model.
7. Growth Theory

Mid-sem → 1 to 4
End-sem → 1 to 7, but 5 to 7 more weightage

Questions will be intuition based, some descriptive based

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LECTURE 1

"Macroeconomics" – overview (bird's eye view) of the economy
'The bigger picture'

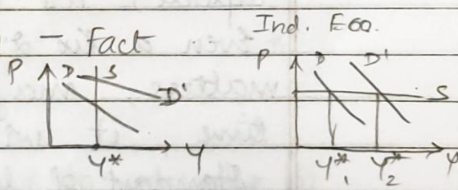
Measurement of productivity of an economy
Output – how much? is it growing?

* Booms & Busts
* Business Cycles
* Exchange Rate
* Inflation } "Evils"
* Unemployment
* Policy
* Budget
* Interest Rate

* Short Run Model
* Long Run Model
* Very Long Run Model.
* Growth Theory
* Goods market
* Labour market } **
* Assets market

Macroeconomics is an applied science
Without policy implications, it is nothing
improving economic behaviour

Growth Rate : 2% to 3%. (Good) – Fact



Demand ∝ Price Level.
(D) . (P)

Supply S , Output Y

"Tractable Essentials" **
Multiplier Effect
Drop in interest rates – ~~expansive~~ expansionary policy.
opp. contraction

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LECTURE 2

Very Long Run Model
* Lies in the domain of Growth Theory
* how an economy performs over a very long period of time.
* US Economy - grew in the last century at the rate of 2-3%.
* Doesn't take into account the fluctuations Average is what matters.
"Business cycle fluctuations" even out over time, hence unimportant.
* Developed economies have much ~~higher~~ lower growth rate as opposed to developing countries.
* Post II World War - Japan's remarkable growth rate (8%) for next few decades. Ghana didn't grow at all (0%)
why? - Long run is also the smaller picture, so we require very long run
* We assume all inputs of production (labor & capital) are employed to the fullest extent.
* Even a ~~bis~~ 2% different growth rate matters, maybe not immediate but in time, it will.

$\boxed{\text{Standard of living matters ~~this~~ a lot in Macroeconomics!}}$

Standard of living - improving it - how?
That is Macroeconomics.

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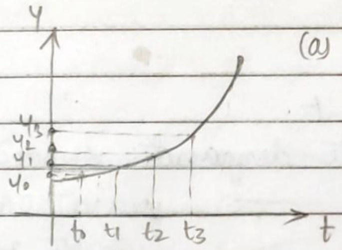
Long Run Model

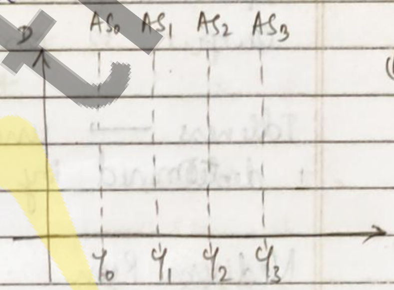
* Inflation, rather, determination of prices & how they fluctuate...

"Aggregate Demand" "Aggregate Supply"

'Given a price level, the maximum amount of output that the economy willing produce'

'Given a price level, the amount of output that ensures simultaneous equilibrium in goods market and service?? market'



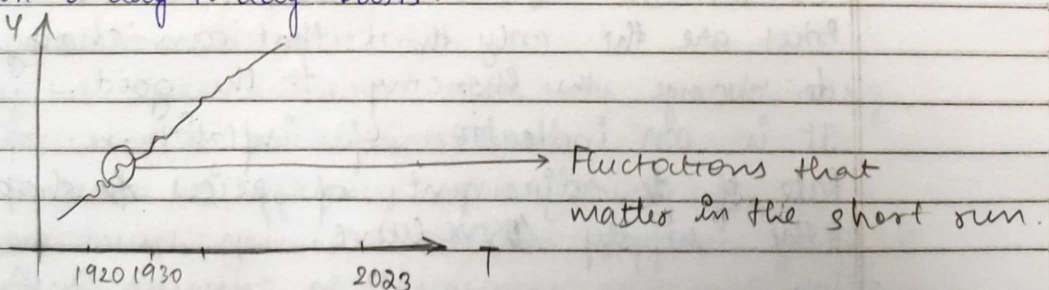


Short Run & Medium Run.

6.1.23 **LECTURE 3**

Short Run

affects policy-makers that have to consider it on a day-to-day basis.



Output doesn't grow in a very state.

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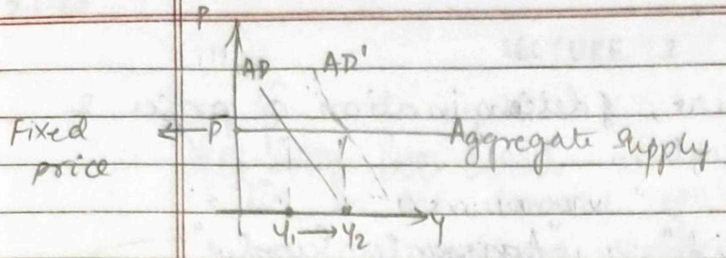
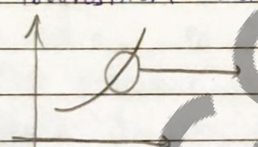


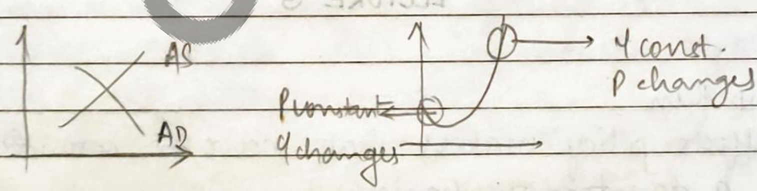
Fluctuations are driven by shifts in the aggregate ~~in~~ demand.
Aggregate demands completely exhausts Aggregate supply,
AD = AS
Equilibrium is attained. Equilibrium Output is produced.

Idleness ⟶ unemployment.
determined by aggregate demand.

Medium Run
Transition between Long Run & Short Run



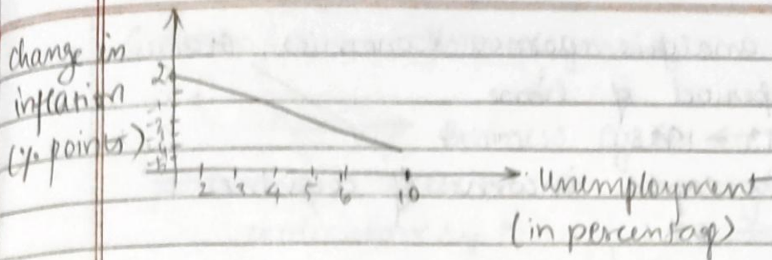
⟶ Upward rising supply curve
Small time period.

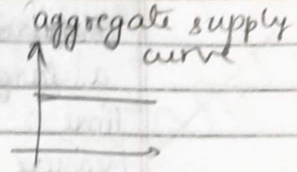


Prices are the only think that can change to change the economy for the good.
It is an indication of inflation.
Rate of ~~de~~ adjusment of prices ~~to~~ shape the supply ~~curve~~ curve.

Price ∝ aggregate demand.

Philip's curve





To reduce unemployment, you must sacrifice & give up inflation.

"Stickiness in prizes"
They take time to adjust.

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LECTURE 4

Growth Theory
AD Theory → GDP
AS Theory
Economic Growth

US Economy, 1960s-2020, Growth Rate : 3.4%.

Q: Why does GDP increase? Why is there growth? Why is Growth Rate positive?

Ans: This implies output is increasing over time
* Available resources actually increase. Why?
Labour, L . Capital, K
available work force, increases because, population increase.
capital stock increases over time, machinary, factories, production units
* Increase in efficiency, productivity.
how much you are able to produce given a fixed amount of resources.
constant input, better and more output

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Comparitive analysis across economies, over a large period of time.

Time : 1913 – 1998

Measure : Per-capita income of countries

* Argentina – 0.7%
* Brazil – 2.3%
* China – 2.4%
* France – 2.1%
* Ghana – 0.1% → absymally low
* India – 1.1%
* Japan – 3.4%
* Spain – 2.0%
* U.K. – 1.6%
* U.S.A – 1.7%

Brazil & Ghana – in 1913 – poor countries

Brazil had healthy growth, Ghana stagnated

They had to deal with inflation, output of Brazil was still increasing, 5 times the output in 1980s as compared to 1913

Ghana had only a 20% increase.

Why contrasting results? – question to policy makes.

→ accumulation & utilization of resources

Thus, we require efficient policy making

growth process has to be smooth, whilst dealing with two undesirable things – unemployment & inflation.

Else, the rich get richer & the poorer get more poor ~~er~~

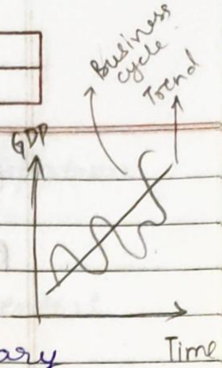
growth cannot be at the cost of the core of the economy for the sake of the top 1%

Growth Unemployment Inflation

Business Cycle

"pattern of recession & recovery"

contractionary ↩ ↪ expansionary

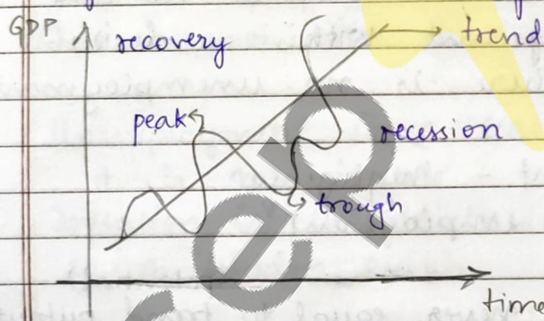


Expansion & Contraction of economic activities along a trend.

Q: Why is the trend important?

Ans: gives an idea wheather output on the whole is increasing or decreasing.

~~tred~~ Trend - no productive unit is sitting idle. The economy is producing upto its fullest potential. No unemployment



peak - "a bubble that is about to burst"

trough - "light at the end of the tunnel"

Everything has a cyclic pattern — (in Macroeconomics) inflation, unemployment, growth, ...

Any measure/ indicator of economic performace never shows a uniformly good or bad stage

Q: What do these fluctuations/gaps measure?

Ans: gives an idea as to what extent as resources are utilized. "Unemployment in resource utilization

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Unemployment & Overemployment shapes & drives the Business Cycle.

Resources are always either under utilized (recession) or overutilized (recovery), never perfectly equal to trend.

Rate of growth of:
o/p in recession < trend < o/p in recovery

Unemployment in a physical sense means the labour force is not working to full capacity. capital is underutilized

From an economists point of view, if people from the labour force are able to find employment within a desirable time period, there is no unemployment

5% unemployment - negligible
"Almost full employment"

Actual output is never equal to trend (potential) output

Q: What about Output Gap? It's implications?
Ans: Output Gap = Actual Output − Potential Output
→ positive - recovery
→ negative - recession

Consumer Price Index - CPI
A basket (fixed) of goods & services that an average consumer uses over a particular period of time, its average price

Increase in CPI (over time) → Inflation in the ~~ec~~ economy

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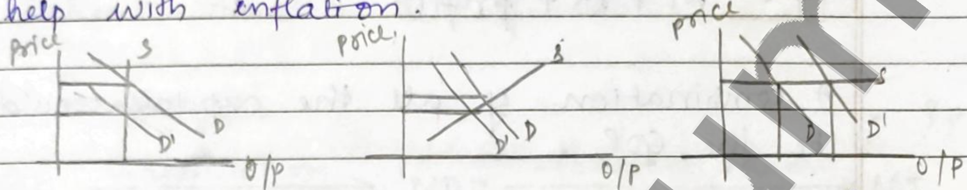
Positive Output gap hints at positive inflation due to rise in aggregate demand.

Unemployment is a scope to increase potential output, which can in some ways help with inflation.



full employment high inflation | mild unemployment less inflation | very high unemployment no inflation.

US - prize of average goods
1920's CPI × 6 ≤ = 2020's CPI
This is not even that high inflation

Unemployment is a leakage in the economy, it is measurable.
Inflation is unpredictable, can't be measured, so a "hidden evil"

See - Philip's Curve
Policy makers trade-off b/w unemployment & inflation.

12.1.23 LECTURE 5

Inflation
Unemployment } Get rid of
& ensure smooth growth
criterias for policy making.

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Total output - $Y = f(L, K)$
L - Labour, K - Capital - 'Factors of Production'
wage rate, w for the labourer
rental rate, r for the capital

$\boxed{Y = wL + rK + \text{profit}}$

?? A combination of all the capital would be the GDP

GDP - Gross Domestic ~~Assets~~ Product
The total income earned from all productive assets of a country, at a point in time, irrespective of their ownership
Collection of all goods & serives of a country

GNP - Gross National Product
Total income earned from all productive assets owned by a country, irrespective of geographical boundaries, they can be anywhere in the world.

GNP = GDP + Income ~~(assets)~~ earned from assets abroad - Income paid to foreign assets operating domestically

$\boxed{\text{GNP} = \text{GDP} + \text{Net property income from abroad}}$

NDP - Net Domestic Product
NDP = GDP - depriciation of capital
depriciation ≡ 11% of GDP - US data

AD:

*
*
*

NI - National Income
Indirect taxes paid by business are taken into account
NI = NDP − indirect taxes
NI = 80% GDP, typically
indirect taxes = 10% of NDP

To summarize,

$$GDP \xrightarrow{-\text{depriciation}} NDP \xrightarrow{-\text{indirect taxes}} NI$$

AD: $\boxed{\cancel{X}^{AD} = C + I + G + NX}$

C → consumption
I → Investments
G → Govt. purchases
NX → Net exports (export - inport)

Ampount spent by domestic agents for foreign goods & services - inport
Amount earned by domestic agents for goods & services from foreign agents - export

Problems in this measurement of Aggregate Demand - AD:
* Value of intermediate goods are ignored, final product is only important
* Considers only currently available goods
* It's a monetary measure, not one of welfare. quantitative, not qualitative

Cyclic ~~fluction~~ fluctuations in AD ≡ creates cyclic fluctuation in business cycle.

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Income Spending Model

* One of the first that we learn, basic
* Cornerstone – mutual interaction between income & expenditure & how they drive the business cycle
* ~~Th~~ Income & expenditure drive each other

$$\text{Spending} \rightleftarrows \text{Output \& income}$$

* Spending ~~it~~ makes another person richer, providing income to another person to spend more

Equilibrium : Total demand = Total supply

$$AD = Y$$

When $AD \neq Y$ – unplanned investment / disinvestment

inventory – increases / decreases

Inventory Unplanned, IU

$$\boxed{IU = Y - AD}$$

$IU > 0$: Additions to the inventory

$IU < 0$: Subtraction to the inventory

IU +ve : Unplanned inventory investment accumulation of excess inventory, cut back on production

IU -ve : Unplanned inventory disinvestment inventory draws down, create more production (produce more) 13.1

C, I, G, NX – components of Aggregate Demand, AD.

Consumption & Income

Lets ignore G & NX for the time being

$\boxed{AD = C + I}$

$\boxed{C = \bar{C} + cY}$, a function of Y (output) (Income)

c – marginal propensity to consume

how much consumption increases to increase by income by 1%.

$\bar{C}$ – some constant consumption

c – what proportion of your increased income goes into expenditure. always a fraction

$0 < c < 1$

consumption α income

If income increase by ₹1, c → constant,

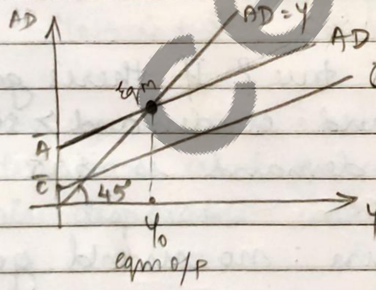
1-c is saved. $\therefore (1-c) > 0$

Saving, $\boxed{S = Y - C}$ (income – consumption)

'Budget constraints'

$\boxed{S = -\bar{C} + (1-c)Y}$ $\qquad (1-c) > 0$

Increase in income ⇒ increase in saving



$\boxed{AD = \bar{A} + cY}$
$\boxed{\bar{A} = \bar{C} + I}$

Y_0 – Equilibrium output

AD–AS DIAGRAM.

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LECTURE 6

Consumption Function : $C = \bar{C} + cY$

"Consumption schedule" "Aggregate Demand Schedule"

"Autonomous Investment"

$\bar{I}$, $\bar{G}$, $\overline{TA}$, $\overline{TR}$

$\overline{TA}$ → Fixed Taxation (Govt. Earns)

$\overline{TR}$ → Fixed Transfer to pvt sector (Govt. spends)

$\bar{G}$ → Govt. spends on infrastructure

$\overline{NX}$ → Import / Export → Net export

Disposable Income, $Y_D = Y - \overline{TA} + \overline{TR}$

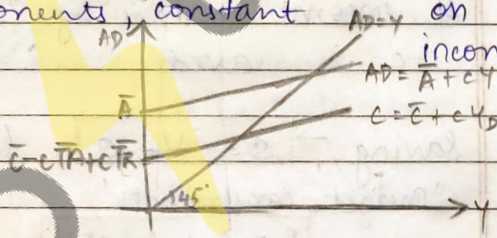
amount for consumption

$C = \bar{C} + cY_D = \bar{C} + c(Y - \overline{TA} + \overline{TR})$

Now, $AD = C + I + G + NX$

$= \underbrace{[\bar{C} - c\overline{TA} + c\overline{TR} + \bar{I} + \bar{G} + \overline{NX}]}_{\text{Autonomous components, constant}} + \underbrace{cY}_{\text{depends on income}}$

$\boxed{AD = \bar{A} + cY}$



"Output produced = Output sold"

$Y = AD$

What if this is not true? If there's gap?

∴ either supply > demand or demand > supply

Equilibrium supply = demand is rare

The notion of equilibrium : no unsold goods

Usually, there remains unsold output.

This output adds to our inventory – inventory accumulation – a part of it is being investment

So, investment goes up.

Equilibrium is hypothetical

The opposite, more demand, would lead to disinvestment due to renting out inventories.

Output that is sold goes into consumption
Output sold $\boxed{Y = C + I}$
We had, $\boxed{Y = C + S}$
So, in this framework, $\boxed{I = Y - C = S}$
Investement is identically equal to savings in a simple economy.

You are saving today to consume tomorrow
'Robinson Crusoe' Economy

~~(Inderin)~~ In a practical sense, Investment is actually an intermediation between a linder and a borrower.
Both benifit.

$$I = S$$
$$Y = C + I + G + NX$$
$$\left.\begin{aligned} Y_D &= Y + TR - TA \\ Y_D &= C + S \end{aligned}\right\} \longrightarrow Y = C + S - TR + TA$$

How much an individual is left with after taxation and gains from tranfers is his disposable income

In an equilibrium
$$C + I + G + NX = C + S - TR + TA$$
$$(S - I) = (G + TR - TA) + NX$$
$$\boxed{S = I + (G + TR - TA) + NX}$$ — Private Savings

'G+TR−TA' − Budget deficit
Net expenditure of the govt. Income
If +ve → Govt. will be dept.

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So, private savings finances investments, budget deficits and channelise loans towards foreigners who deal in import / export

'Real' & 'Nominal' GDP

Inflation, $\boxed{\pi = \frac{P_t - P_{t-1}}{P_{t-1}}}$

$P_t \rightarrow$ Price level of time period t (Today)

$$\boxed{P_t = P_{t-1}(1+\pi)}$$

Price Indices
- → GDP deflator = Nominal GDP / Real GDP
- → CPI
- → PPI

$$\frac{\text{Nominal GDP in the base year}}{\text{Real GDP at base prices}} = \text{GDP deflator}$$

CPI – cost of buying a fixed basket of goods & services of an average urban consumer – a retail measure

PPI – same as CPI, but includes prices from the very first transaction, intermediate products and all

Interest Rate – Real & Nominal

~~Rate~~ 'Real' values take into account inflation. So more accurate

Exchange Rate – Price of foreign currency in terms of domestic currency.

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Fixed Exchange Rate — determined by centralized bank (RBI)
Floating Exchange Rate — determined by market value

Our Exchange Rate is a mix of both.

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LECTURE 7

Output / Income ⇒ National Income

IDENTITY : AD = Output at equilibrium

$AD = C + \bar{I} + \bar{G} + \bar{NX}$ $\quad (\bar{\ }) \Rightarrow$ autonomous / fixed

$C = \bar{C} + cY_D$

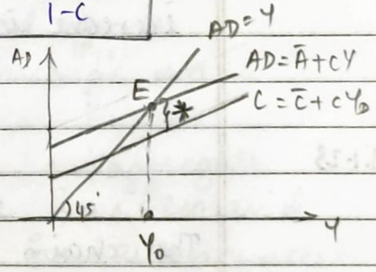
$C = \bar{C} + c(Y + \bar{TR} - \bar{TA}) = \bar{C} - c(\bar{TA} - \bar{TR}) + cY$

Total $Y = \{\bar{C} - c(\bar{TA} - \bar{TR}) + \bar{I} + \bar{G} + \bar{NX}\} + cY$

$\boxed{AD = Y = \bar{A} + cY}$

Y_0, equilibrium output : $\boxed{Y_0 = \frac{1}{1-c}\bar{A}}$

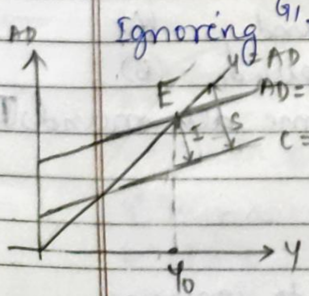
$\boxed{\Delta Y_0 = \frac{1}{1-c}\Delta\bar{A}}$



In a simple economy,
where $AD = C + I$

* That gap b/w AD schedule & C schedule $= \bar{I} + \bar{G} + \bar{NX}$ measure of "planned" investment in a simple economy $(AD = C + I) = I$

Ignoring G, TA, TR, NX — "A" Robinson Crusoe" Economy



At equilibrium, E —
Savings = planned investment
$S = I$

Above Y_0, $S > I$

$Y = C + S$, $\quad AD = C + I$

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If you bring in G, NX, TA, TR –

$Y = C + S + TA - TR$

$AD = C + I + G + NX$ + net imports

At $Y = AD$, $\boxed{I = S + (TA - TR - G) - NX}$

private savings; net revenue earnt by govt (budget surplus) (assuming +ve); net exports

Example: say $\bar{A} = 1$ billion, $c = 0.9 \Rightarrow \frac{1}{1-c} = 10$. $Y_0 = ?$

Answer: $Y_0 = 10$ billion → 10 times

"Multiplier Effect"

Increase in any of the autonomous values that make up AD, increases equilibrium output.

"Increase in Y_0 is more than just one time"

actually 5, 10 times and more

∵ multiplier effect of the economy.

increase in AD < increase in Y_0.

19.1.23 LECTURE 8

The chain reaction of one man's increased expenditure increasing the income of another that overall increases the equilibrium output – "MULTIPLIER EFFECT"

"Keynesian Multiplier"

Round (R)
- → Increase in demand this round (A)
- → Increase in production this round (B)
- → Increase ~~totally~~ in total income all rounds (T)

(Tabular explanation →)

(remember – c : marginal propensity to consume)

R	A	B	T
1	$\Delta\bar{A}$	$\Delta\bar{A}$	$\Delta\bar{A}$
2	$c\Delta\bar{A}$	$c\Delta\bar{A}$	$(1+c)\Delta\bar{A}$
3	$c^2\Delta\bar{A}$	$c^2\Delta\bar{A}$	$(1+c+c^2)\Delta\bar{A}$
4	$c^3\Delta\bar{A}$	$c^3\Delta\bar{A}$	$(1+c+c^2+c^3)\Delta\bar{A}$

We observe a geometric series, which when we sum to infinite rounds —

$T = (1+c+c^2+c^3+\cdots)\Delta\bar{A} \Rightarrow \boxed{T = \frac{1}{1-c}\Delta\bar{A}}$

The added boost in the demand decreases with increase in number of rounds $\{c<1 \Rightarrow 1>c>c^2>c^3\cdots\}$

Increased expenses fall over time

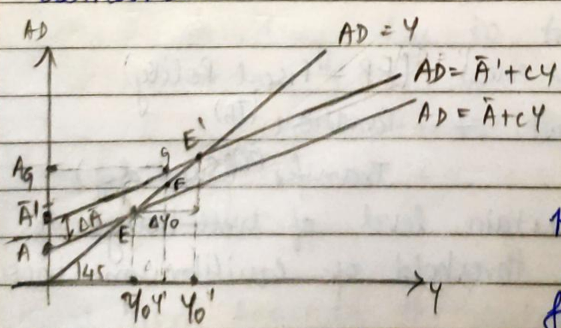
$$\Delta AD = \Delta\bar{A} + c\Delta\bar{A} + c^2\Delta\bar{A} + c^3\Delta\bar{A} + \cdots$$

$$\boxed{\Delta AD = \frac{1}{1-c}\Delta\bar{A}} \quad \boxed{\Delta AD = \alpha\Delta\bar{A}}, \quad \boxed{\alpha = \frac{1}{1-c}}$$

Change (↑se) in Aggregate demand is a multiples of the adhoc change (↑se) in autonomous demand

&

The multiplier is the increase in aggregate demand due to a one time change (increase) in autonomous demand.



A_G – one shot ↑se in agg demand & expenditure due to govt. policies (assume)

How does the gap GF explain the journey from E to E'?

↑ Increase in demand GF decreases as we ~~move~~ move Y' closer to Y_0 & vanishes at E'

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The increased income increases demand, but the ~~the~~ increase in demand decreases as we move toward E'

& $\Delta Y_0 = \frac{1}{1-c} \Delta \bar{A}$, observe the decreasing geometric series $1, c, c^2 \ldots$

The increase happens during a boost, when the economy enter a recession, $\Delta \bar{A}$ can be -ve, i.e., a decrease. It would have a similar negative impact on Y_0. Instead of going up by a multiplier effect, it would go down by a multiplier effect.

For the economy to converge to an equilibrium, always $0 < c < 1$. Marginal propensity must be a fraction.

Coming up next!
How to handle these cyclic fluctuations ~~during~~ while formulating policies?

25.1.23 LECTURE 9

"The Govt. Sector"
Fiscal Policy (a.k.a Govt. Policy) (FP – Fiscal Policy)
Tools of the Govt. – * Taxation (TA)
* Transfer (TR) (Subsidy)
To maintain a certain level of well-being to ensure a certain threshold of equilibrium income

Can create an artificial demand to increase govt. purchases. (G)

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TA & TR — vary as per income bracket of people "per unit"
Else, if same for all "lump sum"

$Y_D = Y - TA + TR$ - Disposible income
$C = \bar{C} + c Y_D$ - Consumption function
influced by govt. policies

Say, govt proposes a proportional taxation "per unit taxation"
income × per unit taxation (always < 1)
$\boxed{TA = tY}$
& if $G = \bar{G}$ & $TR = \overline{TR}$
then $C = \bar{C} + c(1-t)Y + c\overline{TR}$
$\boxed{C = \bar{C} + c\overline{TR} + c(1-t)Y}$
marginal propensity to consume out of (disposible) income : $c(1-t)$

$AD = C + I + G + NX$
$\boxed{AD = [\bar{C} + c\overline{TR} + \bar{I} + \overline{NX} + \bar{G}] + c(1-t)Y}$
$\boxed{AD = \bar{A} + c(1-t)Y}$

Eqm : $Y = AD = \bar{A} + c(1-t)Y$ at Y_0
$\boxed{Y_0 = \dfrac{\bar{A}}{1 - c(1-t)}}$ $\bar{A} = \bar{C} + c\overline{TR} + \bar{I} + \bar{G} + \overline{NX}$

Govt. sector now makes a difference.

Induced spending due to transfers : $c\overline{TR}$
which increases Y
Income tax now lowers the multiplier from $\frac{1}{1-c}$
to $\boxed{\dfrac{1}{1-c(1-t)}}$

Apart from boosting the output, govt. also stabilizes it.

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(mainly during a boost)

Income Tax : "automatic stabilizer"
without direct intervention of the govt., the economy reaches a stable state.
It controls the change in output ΔY_0 when there is a change in AD
ΔY_0 is "kept within limits"
$\because$ t lowers the multiplier that controls ΔY_0
In recovery, it ensure that output doesn't increase beyond control and during a recession, transfers & also acts a stabilizer

The current recession has set in through inflation, an ~~postmath~~ aftermath of the supply shock caused by the pandemic

Recession – Transfers, TR : automatic stabilizer
ensures a certain minimum level of output.

Less income, less expenditure, less output
$\therefore$ less demand – Recession.
Here, govt. creates an "artificial demand" by unemployment benifits acting as automatic stabilizers, to ensure people can consume even with-out a job.
This limits the fluctuations in the business cycles.

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These stabilizers are the reason several economies survived post WWII, as opposed to pre WWII

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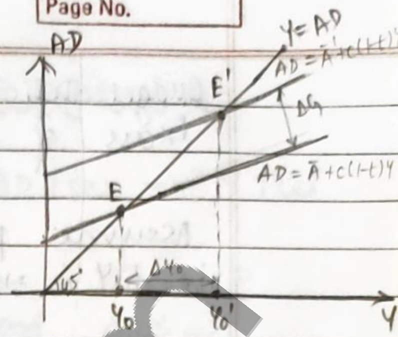

Effect of FP on ΔY_0

Say ΔG is +ve, i.e., G ↑ses

$$Y_0 = \frac{\bar{A}}{1-c(1-t)}$$

$$\boxed{\Delta Y_0 = \frac{\Delta G}{1-c(1-t)} = \alpha_G \Delta G}$$

$$\text{Multiplier} = \frac{1}{1-c(1-t)} = \alpha_G$$

Say, $c = 0.8$, $t = 0.25$
$\alpha_G = 2.5$

Say $\bar{TR}$ ↑ses, the effect is a bit different
Say ~~if~~ there is "lump sum" increase then
$\Delta\bar{TR}$ would show up in ΔA as $c\Delta\bar{TR}$

$$\therefore \boxed{\Delta Y_0 = \frac{c}{1-c(1-t)}\Delta\bar{TR} = c\alpha_G \Delta\bar{TR}}$$

$$\boxed{\Delta Y_0\Big|_{\Delta G} > \Delta Y_0\Big|_{\Delta\bar{TR}}}$$ for certain $|\Delta\bar{TR}| = |\Delta G|$
$\because \alpha_G > c\,\alpha_G \quad \because c<1$

Direct effect : t ↑ses ⇒ Y_D ↓ses, C ↓ses, AD ↓ses
Indirect effect : t ↑ses ⇒ Multiplier is smaller, shocks are absorbed
i.e, shocks induced after change in AD have lesser influence on ΔY_0

Net output depends on which effect dominates the other.

~~1~~1.2.23 **LECTURE 10**

Budget - presented today by Nirmala Sitaraman

Budget Surplus (BS)
Excess of govt. revenue over govt. spending.
$\boxed{BS = TA - \bar{G} - \bar{TR}}$ → $\bar{TR}$: lump sum transfer to citizens
→ $\bar{G}$: amount spent on infrastructure

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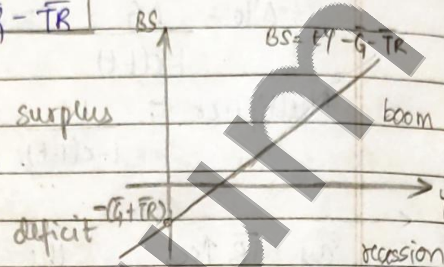
Budget Deficit ($\boxed{BD = -BS}$)
Excess of govt. expenditure over govt. revenue

Assume proportional income taxation
i.e., $t.Y$ $\Rightarrow$ $\boxed{BS = tY - \bar{G} - \bar{TR}}$



low level of income drives BS surplus
high income contributes to BS.

Often, it is not only fiscal policy that determines BS or BD, there are other factors. govt. is not always to blame or praise.

Example: if investors are optimistic about investment, I ↑ses $\Rightarrow$ AD ↑ses $\Rightarrow$ Y ↑ses
Thus, taxation increases (tY ↑ses), which puts us in BS.
Similarly if investors are pessimistic, we may be in BD.

Thus, aggregate / macroeconomic factors matter

Like, if we are in a recession, govt. would have to give unemployment benefits,
$\bar{TR}$ ↑ses $\Rightarrow$ BS ↓ses
But, this in turn will later on ↑se Y
so, the balance will return.

We can try to measure the impact of a fiscal policy on BS by the amount of change in BS due to the policy.

$$\boxed{BS = tY - \bar{G} - \bar{TR}}$$

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Suppose govt. decides to spend more,
$\bar{G}$ ↑ses ⇒ BS ↓ses.
But indirectly, $\bar{G}$ ↑ses ⇒ AD ↑ses ⇒ Y ↑ses
∴ tY ↑ses ⇒ BS ↑ses.
Which of the two possibilities is correct?
Analyse – $\Delta G > 0$ – effect on ΔBS = ?
⇒ $\Delta Y_G = \alpha_G \Delta G$ (induced ↑se in income)
$\Delta TA = t\alpha_G \Delta G$ – increase in govt. tax collection/revenue

$\boxed{\Delta BS = (t\alpha_G - 1)\Delta G}$ $(\Delta TA - \Delta G)$

now, $\alpha_G = \dfrac{1}{1-c(1-t)}$

$$\Delta BS = \left[\frac{t}{1-c(1-t)} - 1\right]\Delta G = \frac{t - \{1-c(1-t)\}}{1-c(1-t)}\Delta G$$

$$\boxed{\Delta BS = \left\{- \frac{(1-c)(1-t)}{1-c(1-t)}\right\}\Delta G}$$ both are fractions c & t.

∴ $\boxed{\Delta BS < 0 \quad \text{if} \quad \Delta G > 0}$

∴ change in govt. expenditure is always higher than change in govt. taxation.

Eg: $c = 0.8$, $t = 0.25$, $\Delta G = ₹1$
$\Delta BS = -₹0.375$
But, the fall is lesser than the gain in ΔG

Naturally, BS ↑ses if t ↑ses
But on the long run, increase in taxation can cause fall in income, so ΔBS can be negative.

Say G ↑ses and TA ↑ses, by the same amount
In order to finance increase in expenditure, the govt. increases taxation. Assume these are lump sum increases.
$\Delta G > 0$, $\Delta TA > 0$ & $\Delta G = \Delta TA$
$BS = TA - G - TR$
⇒ $\underline{\Delta BS = 0}$. There is no effect on BS.

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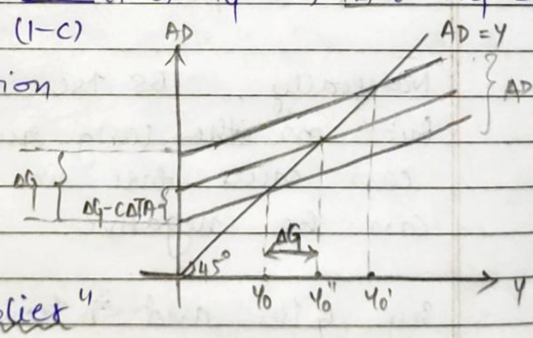
But, what effect would this have on the welfare and income of the people. The relevant question is - what is the impact on Y? $\Delta Y = ?$

ΔG ↑ses ⇒ Y_0 ↑ses
but ΔTA ↑ses ⇒ Y_D ↓ses ⇒ Y_0 ↓ses.
What is the resultant effect on Y_0?

Due to ΔG, $\boxed{\Delta Y_0 = \alpha_G \Delta G}$
TA↑ses ⇒ Y_D ↓ses ⇒ C↓ses ($\because$ $c\Delta TA$)
$\boxed{\Delta Y_0 = -\alpha_G c \Delta TA}$ due to ΔTA
and $\Delta G = \Delta TA > 0$, $c < 1$
$\therefore$ $\boxed{\Delta Y_0 = \alpha_G(1-c)\Delta G}$ (net) Effective change
The amount by which Y_0 falls is lesser than the increase.
$\therefore$ Overall, equilibrium income increases.
(tax multiplier < govt multiplier)
↳ "weaker"

By how much would Y_0 increase?
$\Delta Y_0 = \alpha_G(1-c)\Delta G = \frac{1}{(1-c)}(1-c)\,\Delta G \Rightarrow \boxed{\Delta Y_0 = \Delta G = \Delta TA}$



It's lump sum taxation
So, there is no "t"

This phenomenone is called
"Balanced Buget Multiplier"
This multiplier = 1 when $\Delta G = \Delta TA$

Applications of Balanced Budget Multiplier
— Next Lecture.

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LECTURE 11

Q → $C = 0.8$, $\alpha = 5$

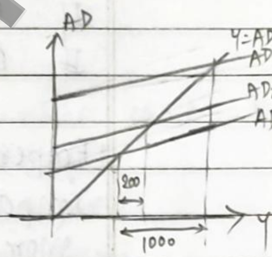
$\Delta G = 200$, $\Delta TA = 200$

$\Delta Y_0 = ?$

Sol^n $\Delta Y_0 = \Delta G = \Delta TA = 200$.

"Balanced Budget Multiplier" "in action"

Either way, expansionary effect on output is more dominant ∵ $c < 1$ (as opposed to contractionary effect)



Increase in output $= 5 \times 200 = 1000$

Decrease in output $= 5 \times 0.8 \times 200 = 800$

∴ $\Delta Y_0 = \cancel{800}\ \underline{200}$

Spending falls by $= c\Delta TA = 160$

If everything else is kept constant,

TR & G ↑ses ⇒ BS ↓ses, Y ↑ses – Expansionary F.P.

t ↑ses ⇒ BS ↑ses, Y ↓ses – Contractionary F.P.

BS is a good indicator of thrust of fiscal policy (Expansionary or Contrationary)

increase in BD or fall in BS doesn't automatically imply expansionary F.P.

↑se in BD ⇏ E.F.P

There are other factors, like boom / recession and so on.

Relative position of the economy on the ~~econ~~ business cycle matters

A measure of fiscal Policy – (thrust of)

Independent of this relative position is

Full Employment Budget surplus : FEBS = BS*

BS at full employment level / potential level of income (Y^* – potential / full employment o/p)

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At full potential
$$\boxed{BS^* = tY^* - \bar{G} - \bar{TR}}$$
dependent on F.P.

BS^* – Standardised / Cyclically Adjusted / Structured / High Employment BS.

actual budget surplus

Actually, $BS = tY - \bar{G} - \bar{TR}$

actual output

$$\boxed{BS^* - BS = t(Y^* - Y)}$$

If $BS^* - BS > 0 \longrightarrow$ recession $\because Y < Y^*$

If $BS^* - BS < 0 \longrightarrow$ boom $\because Y > Y^*$

Empirically, taxation has the biggest impact on fluctuations in BS. 5 times more impactful as compared to changes in autonomous spending

There is no certainty with the value of Y^*. This changes from country to country.

The deviation of BS from full potential BS is a measure of the thrust of F.P. But, it is still not sufficient.

For example, $\left.\begin{array}{l} t \uparrow ses \\ G \uparrow ses \end{array}\right\} \longrightarrow Y \uparrow ses.$

Current level of income is influenced by speculations in future F.P.

Keep in mind, FP depends on several factors – t, TR, G, aggregate shock.

It is hard to measure the path before hand with only one parameter.

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"Money Market Equilibrium"
"Goods Market Equilibrium"
"Simultaneous Equilibrium in Money Market & Goods Market"
IS-LM Model

8.2.23

LECTURE 12

MONEY, INTEREST RATE & INCOME
Till now, we have seen the goods market, now we introduce assests market.
The interaction between these two market that determines equilibrium interest rate and output is what makes up IS-LM model.
We observe how interest rate i.e., monetary policy influences the business cycle and output.

Immediately before recession, ~~IR~~ interest rate (i) is high; during recession rate low and as the economy pick up, interest rate rises.
Why?

Relationship b/w money & output through interest rate.
money ⟶ interest rate ⟶ output.

IS-LM model begins with the simple model that we have studied so far, but we introduce a new variable, interest rate.
Investment is no longer autonomous, but is inversely proportional to interest rate
$I \propto \frac{1}{i}$
$I(i)$ → a function such that I↓ses as i↑ses

$AD = C + I + G + NX$

but, $AD = C + I(i) + G + NX$

What determines i?

<u>Answer</u> — Money Market

A new theoretical framework with simultaneous equilibrium in goods market & Money market.

Y, i

So far fiscal policy worked through the goods market, now it effects the money market. How?

Due to an expansionary F.P., excessive spending results in inflation. To curb the rise in inflation, i increases $\Rightarrow$ I falls $\Rightarrow$ curbs the growth.

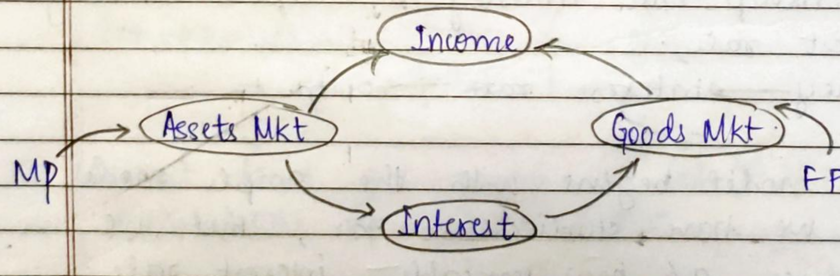
This is a <u>possibility</u>.

Rise in output may be choked.

Assets Market may include Money Market & Bond Market.

FP - Fiscal Policy

MP - Monetary Policy



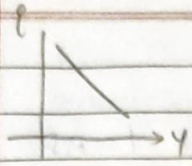
We had one equation, one unknown so far

$C+I+G+NX = AD = Y$, Y

Now we have two unknowns Y, $I(i)$

$\boxed{Y = C + I(i) + G + NX}$, an equilibrium scenario.

A combination of Y & $I(i)$ for which equilibrium is valid.



combination for goods market equilibrium

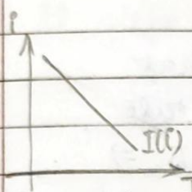
IS - INVESTMENT ~~SUPPLY~~ SAVING CURVE

While talking about IS schedule, we need to be careful of certain things -

$i\uparrow \Rightarrow I\downarrow \Rightarrow Y\downarrow$ $\qquad i \rightarrow I \rightarrow Y$

say $i_1 \longrightarrow Y_1$ unique.

$i_1+\Delta i \longrightarrow Y_1+\Delta Y$ i.e., $\Delta i \xrightarrow{\text{drives}} \Delta Y$



Planned Investment

$\boxed{I = \bar{I} - bi}$

b → slope coefficient

b - very high ⇒ investment is extremely sensitive to interest rate

$\bar{I}$ → intercept coefficient

Now, $AD = C + I + G + NX$

$C = \bar{C} + cY_D = \bar{C} + c(Y + \bar{TR} - tY)$

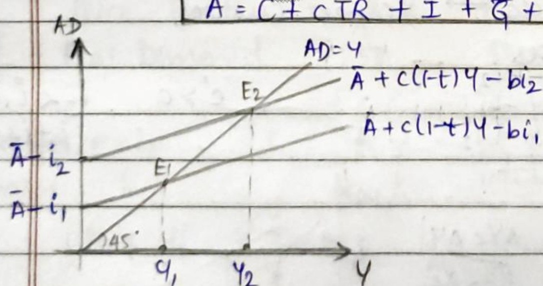
$Y_D = Y + \bar{TR} - TA$, $TA = tY$

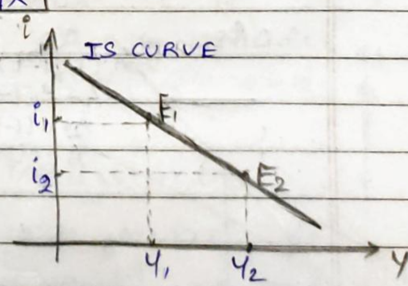
$I = \bar{I} - bi$

$AD = [\bar{C} + c\bar{TR} + c(1-t)Y] + [\bar{I} - bi] + \bar{G} + \bar{NX}$

$\therefore$ $\boxed{AD = \bar{A} + c(1-t)Y - bi}$

$\boxed{\bar{A} = \bar{C} + c\bar{TR} + \bar{I} + \bar{G} + \bar{NX}}$





Say $i_1 \rightarrow i_2$, $\Delta i < 0$ i.e, i falls.

IS CURVE : All possible combinations of (i, Y) that make up $AD = Y$, goods market equilibrium.

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LECTURE 13

Equilibrium is a situation where $Y = AD$

$$\boxed{Y = AD = \bar{A} - bi + c(1-t)Y}$$

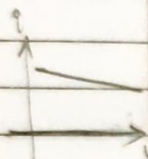
IS CURVE: $\boxed{Y = \alpha_G(\bar{A} - bi)}$, $\boxed{\alpha_G = \dfrac{1}{1 - c(1-t)}}$

$$Y = \alpha_G \bar{A} - b\alpha_G i$$

$$\boxed{\frac{\partial i}{\partial Y} = \frac{-1}{b \cdot \alpha_G}}$$

b – sensitivity parameter

$I = \bar{I} - bi$

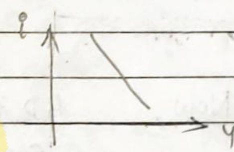


If 'b' is very large, it means that due to a change in interest rate, there is a large change in AD ⇒ there is a large change in Y_0

⇒ Flatter IS schedule

Illy 'b' is small

⇒ Steeper IS schedule



High b ⇒ for Δi, large ΔY_0

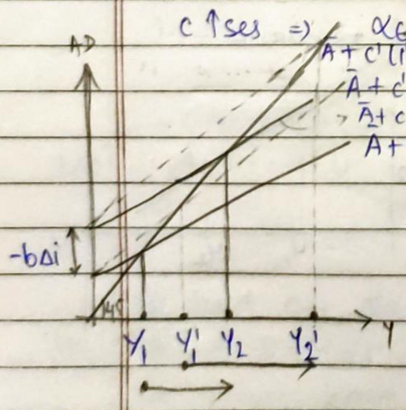
Low b ⇒ for Δi, small ΔY_0

How does multiplier influence $\partial i / \partial Y$?

Answer: large α_G → flatter curve

small α_G → steeper curve

c ↑ses ⇒ α_G ↑ses



$c' > c$, $i_2 < i_1$

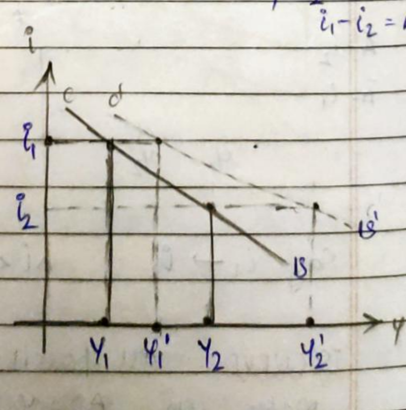
$i_1 - i_2 = \Delta i$

$\Delta Y < \Delta Y'$

$\alpha_G < \alpha_G'$

$c < c'$

For same Δi



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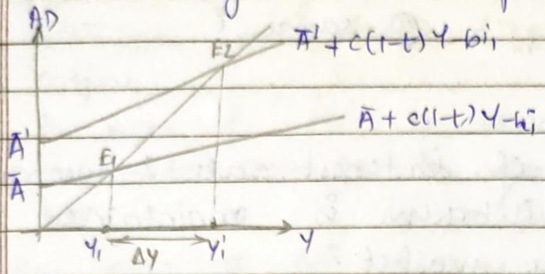
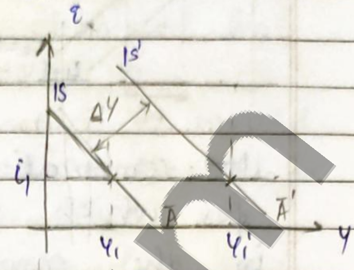
Now, if t ↑ses, α_G ↓ses ⇒ steeper IS
contractory FP ⟶ steeper IS schedule.





$\Delta Y_1 = \alpha_G \Delta \bar{A}$
IS slope doesn't change, only shift occurs

10.2.23

LECTURE 14

~~LIQUIDATED~~ LIQUIDITY MONEY (LM) SCHEDULE

Equilibrium ⇒
Demand for Money = Supply of Money.
We ~~are~~ are concerned with "real demand for money" or "demand for real balances". which is equal to "supply of real money".
To understand this, let us look at two scenarios —

		A	B
$\bar{M}$	Nominal Demand	100	200
P	Price level	2	4
(L) = $\frac{\bar{M}}{P}$	Real Demand	50	50

Price means different things to different people. That's why consider real demand.

Demand for real balances, L

$\boxed{L = kY - hi}$

Y → Income; i → interest rate

L↑ses if Y↑ses : if you earn more, you demand (want) more.
L↓ses if i↑ses : because it's more lucaritive to invest the money & get returns, you spen less.

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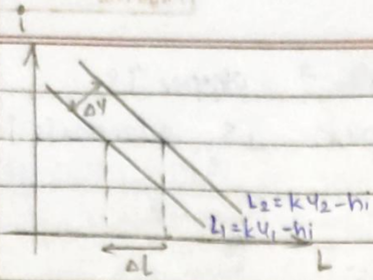


$Y_2 - Y_1 = \Delta Y > 0$

$L_2 - L_1 = \Delta L$

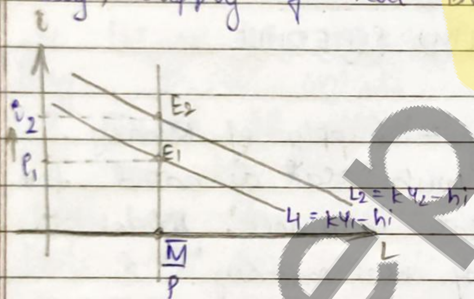
@ same i

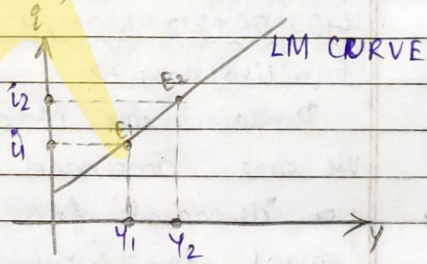
LM → combination of interest rate & income for which equilibrium is maintained in the money market.

To maintain equilibrium, L must be maintained ⇒ i & Y move in the same direction.

Equation for LM curve: $\boxed{i = \frac{1}{h}[kY - L]}$

Say, supply of real balances is fixed. $(\bar{M}/P)$





LM CURVE: $\boxed{i = \frac{1}{h}\left(kY - \frac{\bar{M}}{P}\right)}$

$$\boxed{\frac{di}{dY} = \frac{k}{h}}$$

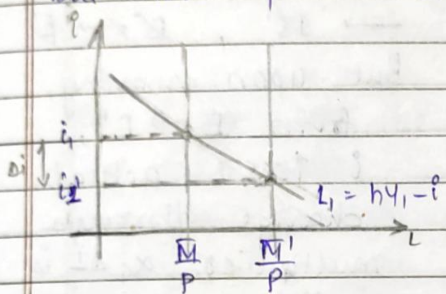
If $h \to \infty$ ⇒ demand for money is very sensitive to i, for a change in i, the change in Y is very large.

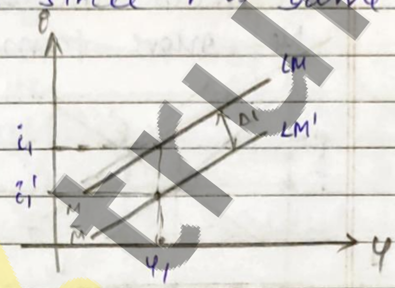
Flatter curve (horizontal)

If $h \to 0$ ⇒ demand for money is not very senstive to i, for a change in i, the change in Y is very small.

Steeper curve (verticle).

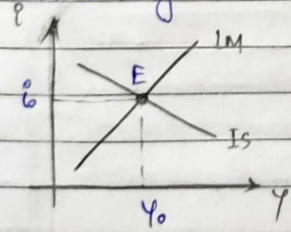
Money supply $\boxed{M^s}$ is held constant till now. If $\bar{M}$ changes, it can change due to change in demand or change in output.
So now if M^s is constant, Y would have to change to accomodate the change in $\bar{M}$.
But now if M^s changes since P is same,



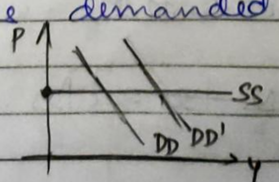


M^s has increased $\because$ $\bar{M}$ has increased
i has decreased to restore equilibrium
The LM shift has moved downward.
but, Y remains the same.

SIMULTANEOUS EQUILIBRIUM
Assumption : Price level P is constant (very short run)
E : Point of simultaneous equilibrium
Combining IL, LM schedules $\rightarrow$ IS-LM curve.



Firms would supply as much as consumers demand at the same price level.
$Y_0 \Rightarrow$ what customers demanded at a fixed price level.



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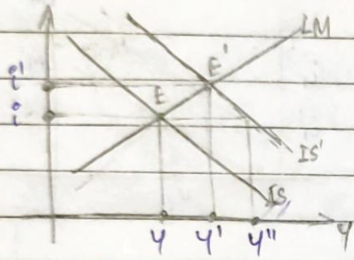
Combined analysis of goods & money market -
we have made certain assumptions.
$\bar{A}$ → remains the same
i.e., we have assumed $\bar{C}, \bar{TR}, \bar{I}, \bar{G}, \bar{NX}$ constant
but, if $\bar{AD}$ ↑ses, due to change in $\bar{I}$↑
(i.e., $\bar{I}$ ↑ses) say due to an FP or if $\bar{G}$ ↑ses.
Y would increase for the same i
We move from IS → IS', E → E'



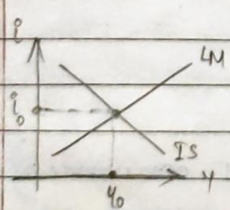
But upon moving from E → E', i ↑ses & actual change through multiplier $\alpha_G \Delta I$ is not seen fully ∵ as i ↑ses it reduces I & full increase in Y is choked.

$Y'' - Y = \alpha_G \Delta I$

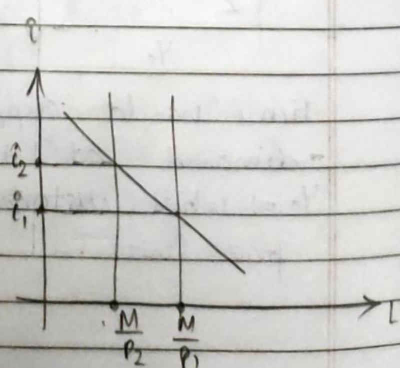
"CROWDING OUT EFFECT"
Initial increase in Y is "crowded out" by an increase in i
So, Net Effect : E → E', i → i', Y → Y'

Next class : P → not constant.

15.2.23 LECTURE 15



Now, if P is not constant,
if P ↑ses
$P_1 \to P_2$, $\frac{M}{P}$ ↓ses ⇒ i ↑ses.



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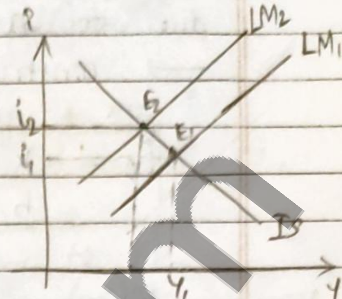
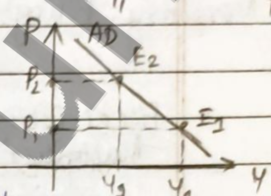
For fixed income, ↑se in i → shift in the LM schedule. Equilibrium Y falls.
IS schedule doesn't change ∵ goods market is unaffected



P vs Y → downward sloping AD schedule



So, if we relax the assumption of fixed prices, IS-LM framework can be used to derive an AD schedule.

$\boxed{Y = \alpha_G(\bar{A} + bi)}$ GOODS MARKET EQUILIBRIUM

→ sensitivity parameter
→ aggregate level of spending
→ multiplier for (if) constant i

$\boxed{i = \frac{1}{h}\left(kY - \frac{\bar{M}}{\bar{P}}\right)}$ MONEY MARKET EQUILIBRIUM

SIMULTANEOUS EQUILIBRIUM

2 Eq^ns, 2 unknowns (i, Y), solving —

$\boxed{Y_0 = \gamma\bar{A} + \gamma\frac{b}{h}\frac{\bar{M}}{\bar{P}}}$

$\boxed{\gamma\,(\text{gamma}) = \frac{\alpha_G}{\left(1 + \frac{k\alpha_G b}{h}\right)}}$

If you consider exogeneous variables relax the fact that $\bar{M}/\bar{P}$ & $\bar{A}$ can change,

Y_0 depends of 2 exogeneous variables — aggregate demand for spending $\bar{A}$ & aggregate supply $(\bar{M}/\bar{P})$

$\bar{A} \rightarrow \bar{C}, \bar{I}, \bar{G}, \bar{TR}$

$\bar{M}/\bar{P} \rightarrow$ inflation, FP or some ~~sth.~~ others.

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In equilibrium – how does Y_0 change due to an exogeneous change in $\bar{M}/\bar{P}$ $(\bar{P})$
$\longrightarrow$ construction of AD schedule

$$\boxed{i_0 = \frac{k}{h}\gamma \bar{A} - \frac{1}{h\alpha_G}\gamma \frac{\bar{M}}{\bar{P}}}$$

Monetary Policy $\xrightarrow{\text{influences}}$ OUTPUT $\xleftarrow{\text{influences}}$ Fiscal Policy

Fiscal Policy Multiplier – how much change in government spending is reflected in Y_0, keeping $\bar{P}$ constant

~~Money~~ Government Multiplier/ (Monetary multiplier) – ?? → Not sure

Fiscal Multiplier how much change in $\bar{P}$ is reflected in Y_0, keeping $\bar{A}$ constant

$$\frac{\partial Y_0}{\partial \bar{G}} = \gamma = \frac{h\alpha_G}{h + kb\alpha_G}$$

k, h, b – sensitivity parameters
α_G – multiplier.

$$\boxed{\frac{\partial Y_0}{\partial \bar{G}} = \frac{\alpha_G}{1 + \frac{kb}{h}\alpha_G}}$$

Now, which multiplier bigger – α_G or γ?
$\boxed{\gamma < \alpha_G}$ – always → $\boxed{\frac{1}{1 + \frac{kb}{h}\alpha_G} < 1}$

Multiplier w.r.t flexible interest rate < Multiplier w.r.t fixed interest rate
∵ crowding out effect

Slope of LM $= \frac{k}{h}$ – sensitivity of money demand to income, interest rate

If $k \uparrow\uparrow$, $h \downarrow\downarrow$
– steeper LM.

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So, for the steep LM, impact of ~~$\frac{dY_0}{dG}$~~ fiscal policy multiplier is negligible.

The more sensitive money policy is to i, there would be an increase on i crowding out.

?? If $h \to 0$, $\gamma \to 0$ less sensitive to FP
$h \to \infty$, $\gamma \to 0$ More sensitive to FP

How is k influence?
If $k \to \infty$, $i \to$ ↑ses. by a high amount to maintain money market equilibrium.
b ↑ses ⇒ I ↓ses ⇒ AD ↓ses ⇒ Y ↓

~~Note~~ If central Band ~~↓~~ decreases i, people take their money back & spend it.
Authorities are setting up a leat without us.
$\bar{M}$ would ↑se ⇒ Y_0 ↑ses.

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LECTURE 16

~~Money~~ Multiplier

Monetary multiplier?

$$\boxed{Y_0 = \gamma\bar{A} + \gamma\frac{b}{h}\frac{\bar{M}}{P}}$$

$$\boxed{\gamma = \frac{\alpha_G}{1 + \frac{kb}{h}\alpha_G}}$$

$$\boxed{\frac{\partial Y_0}{\partial\left(\frac{\bar{M}}{P}\right)} = \frac{b}{h}\gamma = \frac{b\alpha_G}{h + bk\alpha_G}}$$

high b, α_G → Flatter IS $[Y = \alpha_G(\bar{A} - bi)]$
investment is much more sensitive to interest rate

low h, k → Steeper IS, stronger multiplier.
money demand not very sensitive to i, so to reduce $\bar{M}$, large reduction in i → ∵ low h.

* ~~A~~, Effect of large reduction in i
  $\Rightarrow$ High increase in $I \Rightarrow AD \uparrow$ses
  $= Y \uparrow$ses.

$\because$ low $k \rightarrow$ ~~der~~ money demand is less sensitive to change in income to increase demand, high increse in income

Essentially - low b, k, $\Rightarrow$ increased (amplified) effect of multiplier

How to policies impact the two markets?

(FP) Fiscal Policy - initial impact on goods market
(MP) Monetary Policy - initial impact on money market

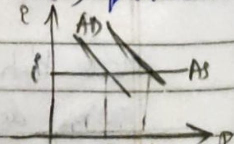
Why do we need policies?
* To maintain & increased growth rate
* Keep inflation under check.

A balance must be maintained ∵
Sometimes, a growth may blow out of proportions when the growth is for a prolonged period & policy makers must be caeful.

We use the IS-LM model to study the impact of any kind of policy.

Both FP & MP impact i & Y
$\rightarrow$ i & Y are detemined by situations in both markets.

Here, we analyse only short run, i.e., price P is fixed.



IS-LM $\rightarrow$ short run.

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Monetary Policy
controlled & managed by the central bank
— India → Reserve Bank of India (RBI)
"Open Market Operation" — OMO
Central banks buys & sells bonds.
open market ~~buy & sell~~ bonds → ploy
to increase money flow in the economy.
~~so~~ Money supply would increase &
govt. bonds would be owned by central Bank
& not by the people. This influences
the price, yeild and interest rate of the
bonds.
open market sale of bonds → public pays
the authority only with money soaks
the money supply in the economy.
& the money stored will be in the
reserve of the central bond.
while buying bonds → (M^s ↑ses)
In order to lure the public hold the
money, to establish equilibrium -
something must be done.
If they want assets → demand for
assets increase → yeild falls → i falls.
If i falls →
public would want of hold the money
⇒ reduced level of interest.
demand for money ↑ses

Money market equilibrium is restored.

say h and k are very high, they would
also impact OMO. Effect of k, h
high h → i ↓ses → I ↑ses.
high k → income ↑ses for AS = AD → Y ↑ses.
Illas to govt multiplier.

$M^s \rightarrow$ money supply

$M^d \rightarrow$ money demand.

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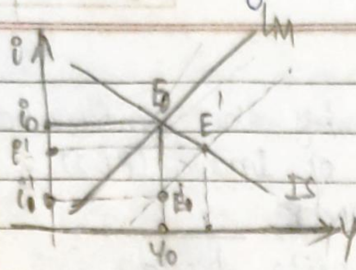


OMO $\rightarrow$ buy bonds.
$\bar{M}$ ↑ses ⇒ Y ↑ses.
Supply for assets had dies → so now demand ↑ses ⇒ supply ↑ses

Now, i ↓ses ⇒ public wants to hold the money.

Ei → money market is same.

For simultaneous equilibrium, Y must increase ⇒

$\boxed{E_0 \rightarrow E_0' \rightarrow E'}$

Richer you are - demand for new-found money demand (M^d)

So, i ↑es → to motivate saving
i ↑ses, M^d ↓se

Equilibrium restored.

17.2.23 **LECTURE 17**

Savings - more lucarative to invest in things that have returns

govt. bonds → sure return

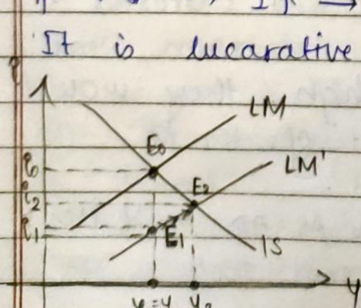
stock - → risk involved

high i ⇒ more lucarative to invest.

other investment opportunities → gold, real estate

if i ↓ → I ↑ → AD ↑ → Y ↑

It is lucarative for companies to borrow @ low i



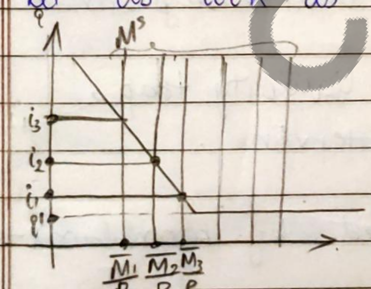
First, there was a drop in i, now due to the increase in Y, ~~we see~~ i ↑ses slightly.

The additional demand is soaked by by increase in interest rate.
This last segment is crucial in icreasing Y
[$E_1 \rightarrow E_2$]
$E_0 \rightarrow E_1$:→ No change in real GDP
∵ Y – no change

Through out this analysis we continuously assume that i has fallen. But what if i cannot fall further? what if in a recession it cannot go any further? Maybe bonds have dried up!
Some situation rate of real interest will go dangerously near zero.
This scenario is called 'LIQUIDITY TRAP'
Central Bank cannot decrease i any further.

Previous recession, consumers lost an interest in investment. Recession was prolonged.

Let us took as supply vs demand schedules.



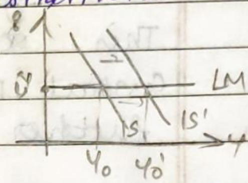
$\overline{M_1} \rightarrow \overline{M_2} \rightarrow \overline{M_3}$ ↑se
$i_1 \rightarrow i_2 \rightarrow i_3$ ↓se
If hits rock bottom can't drop beyond i_1'
i_1' → nominal i

It was a recession, govt. decides to enter OMO.
Further & further still increase in $\overline{M}$
what if they can't go any further?
People can't use loans to invest further.
We now find an elastic demand for money.

No matter how much demand is created public would still hold onto that money that is supplied!

Notice :
During downward slope of M^d_0, i was adjusting ~~fix~~ irrespective of ~~of~~ the supply
But at the lower bound, they cannot and do not have to drop interest rates any further.

Only fiscal Policy can save the day!
IS schedule must shift right! Y would increase simply by increasing government spending



Monetary policy is rendered useless.
OMO are completely ineffective.

"Keynes" – "Keynesian" critique
Classical economics is a ~~moderac~~ says markets will deal with recession.
But, when you enter a liquidity trap, government has to intervene.

Opposite scenario – proposed by monetarists
'CLASSICAL CASE'
$M^d = hi + kY$, $h \to 0 \Rightarrow M^d = kY$
LM curve is unresponsive to any kind of interest rate.
Equilibrium o/p is fixed, can change only with shift of LM curve → due to Monetary Policy. FP is useless
Expansion FP is crowded out by fall in I

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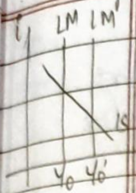
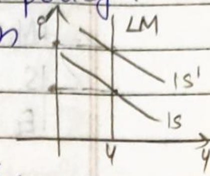


~~Sensitivity~~ ~~of~~ due to monetary policy & fiscal policy ~~of money demand~~ depends on 'h'



Quantity Theory of Money → Next Class.

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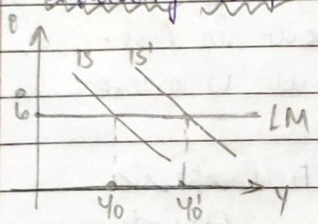
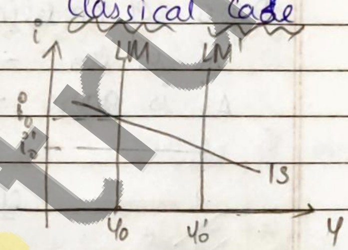
LECTURE 18

Liquidity Trap



Classical Case



CLASSICAL CASE

Nominal GDP is entirely dependent on the quantity of money in circulation.

Completely independent of i

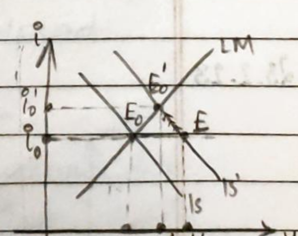
Theory by Monetarists 'Monetary Theory'

Money market forces, no intervention from authorities.

$M^d = kY$ → Real O/P
Real GDP

$\bar{M} = kPY$ → Nominal GDP

Now, $Y = \alpha_G(\bar{A} - bi)$, $\alpha_G = \frac{1}{1-c(1-t)}$



FP → G, t → α_G
→ $\bar{A}$

Say G → increases / TR ↑ses

IS curve — shifts right

Y → increases ⇒ M^d increases, people are richer.

To restore equilibrium in money market, i is increased to control excess demand

I will decrease ⇒ Y decreases ∴ AD decreases.

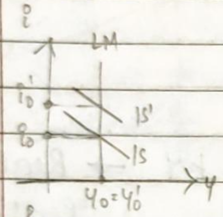
Y is choked off / "crowed out".

Factors that affect degree of crowding out

* LM curve is flatter – Δi_0 is less, ΔY_0 is more.
* IS curve is flatter – Δi_0 is more, ΔY_0 is ~~more~~ less
* E_0 to E – captures the multiplier effect
  $Y = Y_0 + \alpha_G \Delta \bar{A}$

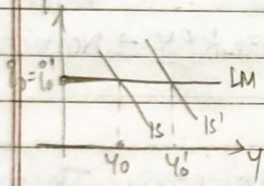
Bottom-line – crowding out always occurs during an expansionary FP

~~Degree of crowding out~~ ($\propto \Delta i$)
Δi is less – crowding out is less
Δi is more – crowding out is more



Classical Case – Full ~~classic~~ Crowding out

Y cannot change, $i \rightarrow$ can increase
FP useless.

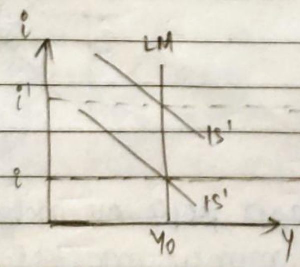


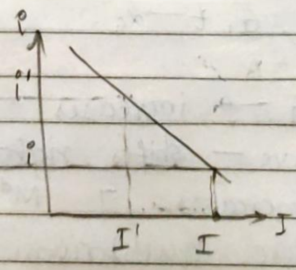
Liquidity Trap – No crowding out

i cannot change, $Y \rightarrow$ fully $\uparrow$ses.
FP fully used.

Implications of Crowding Out – Next Class.

23.2.23 LECTURE 19





Classical Case –
$i \uparrow$ses, $I \downarrow$ses, Y constant

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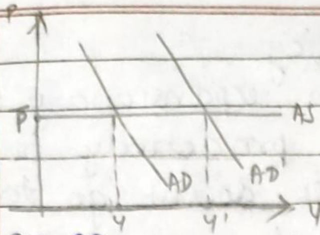
very short run

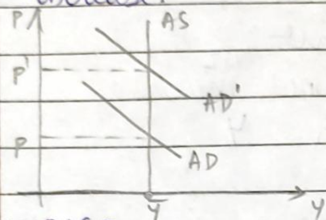


AD ↑ses
Firms ↑ses workforce
Production ↑ses
Y ↑ses
Resources are under-employed.

UNDER-EMPLOYED

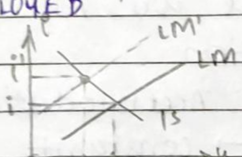
Here, there is potential for output to increase.

very long run



AD ↑ses → Y should ↑se
$\bar{Y}$ is full potential
P ↑ses to curb the increase in AD.
$\frac{\bar{M}}{P}$ ↓ses (supply of money)

FULLY-EMPLOYED



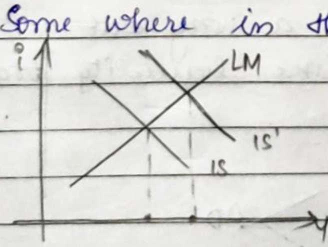
LM schedule shifts leftward.
i ↑ses ⇒ I ↓ses ⇒ AD ↓ses
so Y ↓ses
∴ Increase & decrease balance out.

Similarly, we can understand decrease in AD.

When output is completely crowded out, interest rate stops adjusting.

This happens in the very long run, when resources are fully employed.

Some where in the middle run —



There is scope for Y to ↑se.

$\left.\begin{array}{l}\text{Income}\\ \text{Consumption}\\ \text{Savings}\end{array}\right\}$ ↑se

This way, we can also finance budget deficits with our increased savings.

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Accomodating Monetary Policy

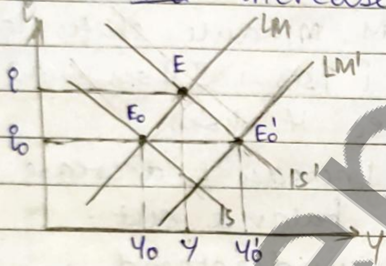
-tive?

As a response to a certain expansionary F.P. Monetary authorities work proactively to ensure that interest rate doesn't go too high and affect investment.
They ensure that I varies within a certain range - Initiative by Central Bank, Short Run scenario.

- Scope for employment
- No crowding out. (not much) NIL

∵ it offsets initial ↑se in Y

A response, now, to increase in AD is to ~~incre~~ increase production.



Fully Accomodative.
i_0 → constant
I, AD → not crowded out.

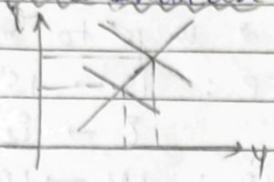
If monetary authories are sufficiently proactive, they would finance BD by buying bonds.
If money supply ↑ses sufficiently, i won't increase too much.

A mix of MP + FP are a way out of tricky situations like liquidity trap & classical case.
They influence →
FP → C, G (, TA, TR)
MP → I (∵ i)
→ AD

MP stimulates inveetive portion of AD, usually only large scale.

MP → affects both money & goods market
FP → strongly affects goods markets.

(Expansionary FP)
Fiscal Increase

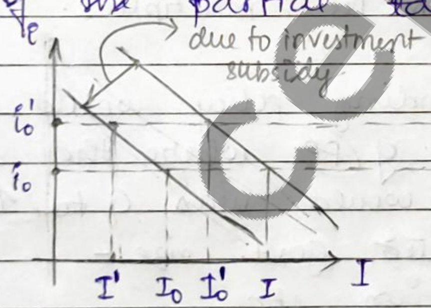


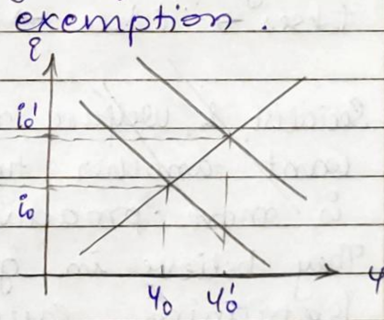
i ↑ses, y ↑ses

(Expansionary MP)
Monetary Increase

i ↓ses, y ↓ses
Better for growth.

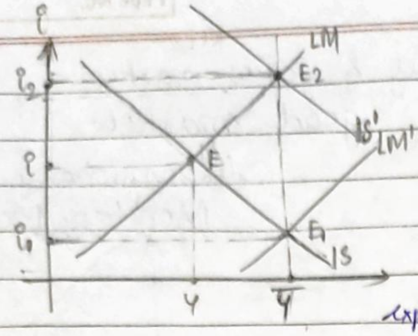
Investment Subsidy – A type of FP
* An initiative by govt. to deal with the undesirable increase in i due to expansionary FPs. They are an incentive to I
* US → Investment Tax Credit
Govt. scheme to maintain a minimum level of investments.
Firms who invest enough get the benefits of the partial tax exemption.





$Y_0 \rightarrow Y_0'$ ↑se.
$i_0 \rightarrow i_0'$ ↑se.
$I_0 \rightarrow I_0'$ ↑se ∵ investment subsidy.

Ideally due to investment subsidy, $I_0 \rightarrow I^*$
but due to $i_0 \rightarrow i_0'$, $I \rightarrow I_0'$
This ensures increase in I



Y – may not be full potential, it can increase to $\bar{Y}$

2 ways to reach $\bar{Y}$

expansionary FP : IS → IS'
$i \to i_2$

expansionary MP : LM → LM'
$i \to i_1$

Which policy to choose?

E_1: Growth minded people, who do not like any negative effect on I would prefer MP.
i ↓ses → I ↑ses → growth is sustainable.

E_2: More conservative people, would want an FP such that t ↓ses now & G ↓ses later when the govt. is doing better.
t ↓ses → C ↑ses → eqm ↑se ∵ multiplier.

Socialist & Welfare minding policy would want another type of FP where the govt. is more proactive would was G to ↑se
They believe in greater govt. size -
1 Job training, education, etc.

So the solution would to create a proper hybrid (mix) of MP and FP.